Dr. Hamoud Aljamaan

Personal Data

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Research Interests

- AI for Software Engineering
- Ensemble Learning
- Software Quality
- Software Security

Work Experience

Associate Professor

Oct 2025 – Now

King Fahd University of Petroleum and Minerals, Saudi Arabia.
Information and Computer Science department
+ SWE 206 Introduction to Software Engineering
+ ICS 557 Advanced Machine Learning

Assistant Professor

Jan 2016 – Oct 2025

King Fahd University of Petroleum and Minerals, Saudi Arabia.
Information and Computer Science department

+ ICS 103 Computer Programming in C
+ SWE 206 Introduction to Software Engineering
+ SWE 311 Principles of Software Engineering
+ SWE 316 Software Design and Architecture
+ SWE 411 Software Engineering Project I
+ SWE 412 Software Engineering Project II
+ ICS 500 Research Methods and Experiment Design in Computing
+ ICS 574 Big Data Analytics
+ ICS 557 Advanced Machine Learning

Chairman

Jan 2019 – Aug 2023

King Fahd University of Petroleum and Minerals, Saudi Arabia.
Information and Computer Science department
Key accomplishments
+ Establishment of the Artificial Intelligence and Machine Learning undergraduate concentration
+ Establishment of the Master of Artificial Intelligence program
+ Establishment of the Cybersecurity and Blockchains undergraduate concentration
+ Establishment of the Master of Cybersecurity programm
+ Establishment of the annual "4IR Data Science Summer School" for Aramco employees since 2019
+ Establishment of the joint Aramco-KFUPM Cybersecurity Chair agreement
+ Establishment of the department female undergraduate and graduate programs
+ Major Revision for the Computer Science undergraduate program
+ Major Revision for the Software Engineering undergraduate program
+ Awarded the ABET re-accreditation for Computer Science undergraduate program
+ Awarded the ABET re-accreditation for Software Engineering undergraduate program
+ Awarded the National Cybersecurity Authority education alignment certificate for the department Cybersecurity masters programs
+ Managed Projects with a total budget exceeding 4 Millions SAR.
+ Recruiting world class new faculty members (8 Graduate Assistants, 8 Assistant Professors, 4 Associate Professors, and 2 Professors)

Summer Consultant

Summer 2018	**Saudi Aramco**, Dhahran, Saudi Arabia. *EXPEC Computer Center / Exploration Applications Services Dept / Applications Technology Division* Review current software development approach and methodology for software development for Exploration, and recommend improvements and best practices in quality, automation, integration and delivery.
Summer 2017	**Saudi Aramco**, Dhahran, Saudi Arabia. *Loss Prevention Department / Risk Assessment Group* SAP Environment, Health, Safety Management (EHSM) module upgrade for the Safety Recommendations System
Summer 2016	**Saudi Aramco**, Dhahran, Saudi Arabia. *EXPEC Computer Center/ Upstream Database Services Division* Predicting Electrical submersible pumps (ESP) failure using machine learning techniques

Full-Time Researcher

Jan 2010 – Dec 2015	**University of Ottawa**, Canada. *Complexity Reduction in Software Engineering (CRuiSE) group* Member of the CRuiSE research group under the supervision of Prof. Timothy Lethbridge. Conducted research on textual modeling languages and contributed to the development on the Umple language adopting these key practises: + Model-Driven Development (MDD) + Test-Driven Development (TDD) + Agile Development + Open source using svn/git

Lecturer

July 2009 – Dec 2015	**King Fahd University of Petroleum and Minerals**, Saudi Arabia. *Information and Computer Science department* Prepared and given Lectures to undergraduate students on programming principles using Java. + ICS 102 Introduction to Computing I

Teaching Assistant

Sept 2012 – Dec 2014	**University of Ottawa**, Canada. *School of Electrical Engineering and Computer Science* Prepared and Supervised laboratories. Marked quizzes and assignments. Provided feedback on students projects. Proctored exams. + ITI1120 Introduction to Computing I - Fall 2012 / Fall 2013 + ITI1121 Introduction to Computing II - Winter 2013 / Summer 2014 + SEG2105 Introduction to Software Engineering - Fall 2014 + SEG2106 Software Construction - Winter 2013 / Summer 2014

Graduate Assistant

Nov 2006 – June 2009	**King Fahd University of Petroleum and Minerals**, Saudi Arabia. *Information and Computer Science department* Prepared and Supervised laboratories. Marked quizzes and assignments. Provided feedback on students projects. Proctored exams. + ICS 102 Introduction to Computing I + ICS 103 Computer Programming in C

Summer Intern

Summer 2005	**Saudi Aramco**, Dhahran, Saudi Arabia. *Integrated Solution Service Department (ISSD)* Developed and enhanced an image processing program in the ISSD.

Education

December 2015	**Doctor of Philosophy** in Computer Science	Emphasis: Software Engineering **University of Ottawa**, Ottawa, Canada Supervisor: Prof. Timothy Lethbridge Dissertation: "*Model-Oriented Tracing Language*: Producing Execution Traces from Tracepoints Injected into Code Generated from UML Models" [pdf]
June 2009	**Master of Science** in Computer Science	Emphasis: Software Engineering **King Fahd University of Petroleum and Minerals**, Dhahran, Saudi Arabia Supervisor: Dr. Mahmoud Elish Thesis: "Software Quality Assessment using Ensemble Models" [pdf]
June 2006	**Bachelor of Science** in Computer Science **King Saud University**, Riyadh, Saudi Arabia Project supervisor: Dr. Mohsen Denguir Graduation project: "Electronic Ordering System Using Hand-held Devices"

Publications

Journals

[J24] M. Alshabrawi, N. Alomari, M. Abushwarib, **Hamoud Aljamaan**. "Text Chunking Strategies for Retrieval-Augmented Generation: A Survey." *IEEE Access*, accepted/in press (2026). DOI: https://doi.org/10.1109/ACCESS.2026.3742213

[J23] R. Baamer, **Hamoud Aljamaan**. "Ensemble deep learning models for stock price forecasting: evidence from the Saudi markets." *PeerJ Computer Science*, 12, Article e4064 (2026). DOI: https://doi.org/10.7717/peerj-cs.4064

[J22] R. Baamer, **Hamoud Aljamaan**. "Enhancing Stock Price Forecasting with Deep Learning: Insights from the Saudi Stock Market." *Computational Economics*, 68(1), pp. 61–109 (2026). DOI: https://doi.org/10.1007/s10614-025-11033-9

[J21] S. J. Alsunaidi, **Hamoud Aljamaan**. "Enhanced Phishing Transactions Detection on Ethereum Network with Tree-based Ensembles: An Empirical Study." *Blockchain: Research and Applications*, Article 100506 (2026). Advance online publication. DOI: https://doi.org/10.1016/j.bcra.2026.100506

[J20] S. J. Alsunaidi, **Hamoud Aljamaan**, M. Hammoudeh. "Leveraging Machine Learning Models to Improve Smart Contract Security: A Survey of Vulnerabilities and Detection Methods." *ACM Computing Surveys*, 58(6), Article 151 (2026). DOI: https://doi.org/10.1145/3772367

[J19] S. J. Alsunaidi, **Hamoud Aljamaan**, M. Hammoudeh. "DIVE: A Multi-Label Smart Contract Vulnerability Dataset." *Scientific Data*, 13, Article 664 (2026). DOI: https://doi.org/10.1038/s41597-026-07025-5

[J18] A. Alazba, **Hamoud Aljamaan**, M. Alshayeb. "SmellyBot: An AI-Powered Software Bot for Code Smell Detection." *Software: Practice and Experience*, 55(10), pp. 1726–1742 (2025). DOI: https://doi.org/10.1002/spe.70006

[J17] R. Sandouka, **Hamoud Aljamaan**. "Cross-Language Code Smell Detection via Transfer Learning." *Applied Sciences*, 15(17), Article 9293 (2025). DOI: https://doi.org/10.3390/app15179293

[J16] N. Alomari, A. Alazba, **Hamoud Aljamaan**, M. Alshayeb. "SmellyCode++: Multi-Label Dataset for Code Smell Detection." *Scientific Data*, 12, Article 1207 (2025). DOI: https://doi.org/10.1038/s41597-025-05465-z

[J15] R. Sandouka, **Hamoud Aljamaan**. "Enhancing Python Code Smell Detection with Heterogeneous Ensembles." *International Journal of Software Engineering and Knowledge Engineering*, 35(7), pp. 963–986 (2025). DOI: https://doi.org/10.1142/S0218194025500287

[J14] S. J. Alsunaidi, **Hamoud Aljamaan**, M. Hammoudeh. "MultiTagging: A Vulnerable Smart Contract Labeling and Evaluation Framework." *Electronics*, 13(23), Article 4616 (2024). DOI: https://doi.org/10.3390/electronics13234616

[J13] **Hamoud Aljamaan**. "Dynamic stacking ensemble for cross-language code smell detection." *PeerJ Computer Science*, 10, Article e2254 (2024). DOI: https://doi.org/10.7717/peerj-cs.2254

[J12] A. Alazba, **Hamoud Aljamaan**, M. Alshayeb. "CoRT: Transformer-based code representations with self-supervision by predicting reserved words for code smell detection." *Empirical Software Engineering*, 29(3), Article 59 (2024). DOI: https://doi.org/10.1007/s10664-024-10445-9

[J11] A. Alazba, **Hamoud Aljamaan**, M. Alshayeb. "Automated detection of class diagram smells using self-supervised learning." *Automated Software Engineering*, 31(1), Article 29 (2024). DOI: https://doi.org/10.1007/s10515-024-00429-w

[J10] A. Alqarni, **Hamoud Aljamaan**. "Leveraging Ensemble Learning with Generative Adversarial Networks for Imbalanced Software Defects Prediction." *Applied Sciences*, 13(24), Article 13319 (2023). DOI: https://doi.org/10.3390/app132413319

[J9] N. Alturayeif, **Hamoud Aljamaan**, J. Hassine. "An automated approach to aspect-based sentiment analysis of apps reviews using machine and deep learning." *Automated Software Engineering*, 30(2), Article 30 (2023). DOI: https://doi.org/10.1007/s10515-023-00397-7

[J8] R. Sandouka, **Hamoud Aljamaan**. "Python code smells detection using conventional machine learning models." *PeerJ Computer Science*, 9, Article e1370 (2023). DOI: https://doi.org/10.7717/peerj-cs.1370

[J7] A. Alazba, **Hamoud Aljamaan**, M. Alshayeb. "Deep learning approaches for bad smell detection: a systematic literature review." *Empirical Software Engineering*, 28(3), Article 77 (2023). DOI: https://doi.org/10.1007/s10664-023-10312-z

[J6] A. Alazba, **Hamoud Aljamaan**. "Software Defect Prediction Using Stacking Generalization of Optimized Tree-Based Ensembles." *Applied Sciences*, 12(9), Article 4577 (2022). DOI: https://doi.org/10.3390/app12094577

[J5] K. E. Hoque, **Hamoud Aljamaan**. "Impact of Hyperparameter Tuning on Machine Learning Models in Stock Price Forecasting." *IEEE Access*, 9, pp. 163815–163830 (2021). DOI: https://doi.org/10.1109/ACCESS.2021.3134138

[J4] A. Alazba, **Hamoud Aljamaan**. "Code smell detection using feature selection and stacking ensemble: An empirical investigation." *Information and Software Technology*, 138, Article 106648 (2021). DOI: https://doi.org/10.1016/j.infsof.2021.106648

[J3] T. C. Lethbridge, A. Forward, O. Badreddin, D. Brestovansky, M. Garzon, **Hamoud Aljamaan**, S. Eid, A. Husseini Orabi, M. Husseini Orabi, V. Abdelzad, O. Adesina, A. Alghamdi, A. Algablan, A. Zakariapour. "Umple: Model-driven development for open source and education." *Science of Computer Programming*, 208, Article 102665 (2021). DOI: https://doi.org/10.1016/j.scico.2021.102665

[J2] A. Al-Shaaby, **Hamoud Aljamaan**, M. Alshayeb. "Bad Smell Detection Using Machine Learning Techniques: A Systematic Literature Review." *Arabian Journal for Science and Engineering*, 45(4), pp. 2341–2369 (2020). DOI: https://doi.org/10.1007/s13369-019-04311-w

[J1] M. O. Elish, **Hamoud Aljamaan**, I. Ahmad. "Three empirical studies on predicting software maintainability using ensemble methods." *Soft Computing*, 19(9), pp. 2511–2524 (2015). DOI: https://doi.org/10.1007/s00500-014-1576-2

Conferences

[C17] N. Alsunaidi, S. Aljbali, Y. Yasin, **Hamoud Aljamaan**. "Arabic Cyberbullying Detection Using Machine Learning: State of the Art Survey." *27th International Conference on Evaluation and Assessment in Software Engineering*, Oulu, Finland, pp. 499–504 (2023). DOI: https://doi.org/10.1145/3593434.3593968

[C16] D. Alomari, F. Anis, M. Alabdullatif, **Hamoud Aljamaan**. "A Survey on Botnets Attack Detection Utilizing Machine and Deep Learning Models." *27th International Conference on Evaluation and Assessment in Software Engineering*, Oulu, Finland, pp. 493–498 (2023). DOI: https://doi.org/10.1145/3593434.3593967

[C15] S. J. Alsunaidi, R. T. Alraddadi, **Hamoud Aljamaan**. "Twitter Spam Accounts Detection Using Machine Learning Models." *2022 14th International Conference on Computational Intelligence and Communication Networks (CICN)*, Al-Khobar, Saudi Arabia, pp. 525–531 (2022). DOI: https://doi.org/10.1109/CICN56167.2022.10008339

[C14] **Hamoud Aljamaan**. "Voting Heterogeneous Ensemble for Code Smell Detection." *2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA)*, Pasadena, CA, USA, pp. 897–902 (2021). DOI: https://doi.org/10.1109/ICMLA52953.2021.00148

[C13] N. Alturaief, **Hamoud Aljamaan**, M. Baslyman. "AWARE: Aspect-Based Sentiment Analysis Dataset of Apps Reviews for Requirements Elicitation." *2021 36th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW)*, Melbourne, Australia, pp. 211–218 (2021). DOI: https://doi.org/10.1109/ASEW52652.2021.00049

[C12] **Hamoud Aljamaan**, A. Alazba. "Software defect prediction using tree-based ensembles." *16th ACM International Conference on Predictive Models and Data Analytics in Software Engineering*, Virtual conference, pp. 1–10 (2020). DOI: https://doi.org/10.1145/3416508.3417114

[C11] **Hamoud Aljamaan**, T. C. Lethbridge, M. A. Garzón. "MOTL: a textual language for trace specification of state machines and associations." *25th Annual International Conference on Computer Science and Software Engineering (CASCON 2015)*, Markham, Canada, pp. 101–110 (2015). [proceedings]

[C10] **Hamoud Aljamaan**, T. Lethbridge, M. Garzón, A. Forward. "UmpleRun: a Dynamic Analysis Tool for Textually Modeled State Machines using Umple." *1st International Workshop on Executable Modeling (EXE 2015)*, *CEUR Workshop Proceedings*, vol. 1560, Ottawa, Canada, pp. 16–20 (2015). [proceedings]

[C9] M. A. Garzon, **Hamoud Aljamaan**, T. C. Lethbridge. "Umple: A framework for Model Driven Development of Object-Oriented Systems." *2015 IEEE 22nd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*, Montreal, QC, Canada, pp. 494–498 (2015). DOI: https://doi.org/10.1109/SANER.2015.7081863

[C8] M. A. Garzón, T. C. Lethbridge, **Hamoud Aljamaan**, O. Badreddin. "Reverse engineering of object-oriented code into Umple using an incremental and rule-based approach." *24th Annual International Conference on Computer Science and Software Engineering (CASCON 2014)*, Toronto, Canada, pp. 91–105 (2014). [proceedings]

[C7] V. Abdelzad, **Hamoud Aljamaan**, O. Adesina, M. A. Garzon, T. C. Lethbridge. "A Model-Driven Solution for Financial Data Representation Expressed in FIXML." *7th Transformation Tool Contest (TTC 2014), co-located with STAF 2014*, *CEUR Workshop Proceedings*, vol. 1305, York, United Kingdom, pp. 65–70 (2014). [proceedings]

[C6] **Hamoud Aljamaan**, T. C. Lethbridge, O. Badreddin, G. Guest, A. Forward. "Specifying Trace Directives for UML Attributes and State Machines." *2nd International Conference on Model-Driven Engineering and Software Development*, Lisbon, Portugal, pp. 79–86 (2014). DOI: https://doi.org/10.5220/0004711500790086

[C5] O. Badreddin, T. C. Lethbridge, A. Forward, M. Elaasar, **Hamoud Aljamaan**, M. A. Garzon. "Enhanced Code Generation from UML Composite State Machines." *2nd International Conference on Model-Driven Engineering and Software Development*, Lisbon, Portugal, pp. 235–245 (2014). DOI: https://doi.org/10.5220/0004699602350245

[C4] **Hamoud Aljamaan**, M. O. Elish, I. Ahmad. "An Ensemble of Computational Intelligence Models for Software Maintenance Effort Prediction." *12th International Work-Conference on Artificial Neural Networks (IWANN 2013)*, *Lecture Notes in Computer Science*, vol. 7902, Puerto de la Cruz, Tenerife, Spain, pp. 592–603 (2013). DOI: https://doi.org/10.1007/978-3-642-38679-4_60

[C3] **Hamoud Aljamaan**, T. C. Lethbridge. "Towards Tracing at the Model Level." *2012 19th Working Conference on Reverse Engineering*, Kingston, ON, Canada, pp. 495–498 (2012). DOI: https://doi.org/10.1109/WCRE.2012.59

[C2] A. A. Bahjat, **Hamoud Aljamaan**, M. Al-Shayeb. "SQL-Guard Design Pattern." *18th International Conference on Software Engineering and Data Engineering (SEDE 2009)*, Las Vegas, NV, USA, pp. 144–148 (2009). [proceedings]

[C1] **Hamoud I. Aljamaan**, M. O. Elish. "An empirical study of bagging and boosting ensembles for identifying faulty classes in object-oriented software." *2009 IEEE Symposium on Computational Intelligence and Data Mining*, Nashville, TN, USA, pp. 187–194 (2009). DOI: https://doi.org/10.1109/CIDM.2009.4938648

Patent

[P1] A. A. Alazba, **Hamoud Ibrahim Aljamaan**, M. R. Alshayeb. "System and method for code smell detection using transformer-based code representations with self-supervision by predicting reserved words." U.S. Patent US 12,591,499 B2; filed 31 January 2024; granted 31 March 2026. Assignee: King Fahd University of Petroleum and Minerals. [patent]

Research Funding

2024 - 2025 | **King Fahd University of Petroleum and Minerals (KFUPM)**
Internal Research Project **Status:** Completed
Title: Leveraging Transfer Learning for Code Smell Detection Across Different Programming Languages
Role: Principal Investigator (PI) **Project number:** INFE2410
Duration: 1 year (extended) **Budget:** SAR 46,000

2020 - 2024 | **Saudi Aramco**
External Project
Title: Researching Upstream Challenges through 4IR Summer School
Role: Cost Center Manager, Co-I **Project number:** ICS-2506
Duration: 4 years **Budget:** Approx. SAR 2.5M

2019 | **Saudi Aramco**
External Project
Title: Studying the feasibility of establishing 4IR Summer School
Role: Cost Center Manager, Co-I **Project number:** ICS-2468
Duration: 1 year **Budget:** Approx. SAR 560,000

2018 | **King Fahd University of Petroleum and Minerals (KFUPM)**
START UP RESEARCH GRANT
Title: Software Maintainability Prediction using Hybrid Data Mining Prediction Models.
Role: PI **Project number:** SR 171014
Duration: 11 months **Budget:** SAR 60,500

2008 | **King Abdulaziz City for Science and Technology (KACST)**
MASTER THESIS FUNDING
Title: Software Quality Assessment using Ensemble Models.
Role: PI **Project number:** GSP-17-132
Duration: 1 year **Budget:** SAR 50,000

THESIS SUPERVISION

- **[Ongoing]** Dalal Aldowaihi, PhD in Computer Science.
- **[Ongoing]** Aishah Altamimi, PhD in Computer Science.
- **[Ongoing]** Yasmin Yasin, PhD in Computer Science.
- **[Ongoing]** Raghad Alzahrani, Msc in Computer Science.
- **[Ongoing]** Norah Alsunaidi, Msc in Computer Science.
- **[Completed]** Shikah Alsunaidi, PhD in Computer Science, Dec 2025.
  - Thesis title: "Lifecycle-Aware and Drift-Resilient Machine Learning for Smart Contract Vulnerability Detection".
- **[Completed]** Rana Sandouka, Msc in Computer Science, Nov 2025.
  - Thesis title: "Transfer Learning Approaches for Cross-Language Code Smell Detection".
- **[Completed]** Rana Baamer, Msc in Computer Science, Sept 2025.
  - Thesis title: "Enhanced Saudi Stock Prices Forecasting with Technical Indicators and Deep Learning Ensembles".
- **[Completed]** Amal Alazba, PhD in Computer Science, Dec 2023.
  - Thesis title: "A Framework for Software Code and Model Smell Detection through Self Supervised Learning".
- **[Completed]** Amani Alqarni, Msc in Computer Science, Dec 2023.
  - Thesis title: "Leveraging Ensemble Learning with Generative Adversarial Networks in Imbalanced Software Defects Prediction".
- **[Completed]** Lama Albakhat, Msc in Computer Science, May 2022.
  - Thesis title: "Robustness And Stability Analyses Of Ensemble Learning Models In Android Malware Detection".
- **[Completed]** Nouf Alturayeif, Msc in Computer Science, Dec 2021.
  - Thesis title: "Aspect-Based Sentiment Analysis of Apps Reviews Using Supervised Machine Learning".
- **[Completed]** Kazi Ekramul Hoque, Msc in Computer Science, Dec 2021.
  - Thesis title: "Hyperparameter Tuning Impact on Machine Learning Forecasting Performance within the Saudi Stock Market".
- **[Completed]** Ahmed Alshaaby, Msc in Software Engineering, Dec 2019.
  - Thesis title: "Software Bad Smells Prediction Using Advanced Machine Learning Techniques".

Projects

Completed | **Umple: a model oriented programming language**
CRuiSE, University of Ottawa, Canada
Open-source language that integrates UML with Java and PHP, supports standalone modeling and embedded code, and generates code from class and state diagrams.

Completed | **MOTL: a model oriented tracing language**
CRuiSE, University of Ottawa, Canada
Model-level tracing language integrated with Umple for UML associations, attributes, and state machines, with constraints to control tracing scope.

2008 | **School Information System**
Research Institute, KFUPM, Saudi Arabia
Requested by *Ministry of Education, Saudi Arabia*
Team member: developed the software requirements specification (SRS) for online school management and teacher affairs systems.

Awards and Honors

- Recipient of the **Research Excellence Award for Journal Publications**, awarded by the Interdisciplinary Research Center for Finance and Digital Economy (IRC-FDE), King Fahd University of Petroleum and Minerals (KFUPM), 2025.
- Recipient of the **Distinction Award for Excellence in Academic Advising**, King Fahd University of Petroleum and Minerals (KFUPM), 2025.
- Supervised the team awarded the **Best Software Engineering Senior Design Project Award** for the project "Qulab" at the Senior Design Project (SDP) Expo, King Fahd University of Petroleum and Minerals (KFUPM), 2023. *Role: Supervisor*
- Led the KFUPM delegation that achieved **Second Place among 51 teams** at the Gulf Programming Competition, Muscat, Oman, 2022. *Role: Delegation Head*
- Received the **National Cybersecurity Authority (NCA) Education Alignment Certificate** for the Cybersecurity Master's programs at KFUPM, 2022. *Role: Chairman*
- Successfully led the **ABET re-accreditation** process for the Bachelor of Science program in Computer Science, KFUPM, 2021. *Role: Chairman*
- Successfully led the **ABET re-accreditation** process for the Bachelor of Science program in Software Engineering, KFUPM, 2021. *Role: Chairman*
- Recipient of the **University Service Award** in recognition of ten years of service at King Fahd University of Petroleum and Minerals (KFUPM), 2016.
- Awarded multiple **University of Ottawa Travel Grants** to support research collaboration and conference participation, including: École Polytechnique de Montréal, Canada (2010–2012); CSER Conference, Victoria, Canada (2012); MODELSWARD Conference, Lisbon, Portugal (2014).
- Recipient of multiple **Saudi Arabian Cultural Bureau (SACB) Excellence Awards**: Fall 2012, Winter 2014, Winter 2015, and Fall 2015.
- Awarded a fully funded **PhD Scholarship** by the Ministry of Higher Education, Saudi Arabia, 2010.
- Recipient of the **Saudi Telecom Company (STC) Distinguished Graduate Award** for ranking among the top ten graduating students at King Fahd University of Petroleum and Minerals (KFUPM), 2008.
- Recipient of the **Outstanding Graduate Academic Performance Award**, King Fahd University of Petroleum and Minerals (KFUPM), 2008.
- Graduated with **Second Class Honors** from King Saud University, 2006.

Professional Service and Leadership

- **Organizing Co-chair and Proceedings Chair** — International Workshop on Security Assurance for AI-Assisted Software Engineering (ASSURE-AI 2027), co-located with SANER 2027, Richmond, Virginia, USA, 9 March 2027 (upcoming).
- **Program Committee Member** — IEEE International Conference on Computational Intelligence and Communication Networks (CICN), 2025–2027 (17th–19th editions; 2027 upcoming).
- **Judge** — Mawhiba, Riyadh, Saudi Arabia, 2019–present. Member of the judging committee for the National Olympiad for Scientific Creativity "Ibda'a". Area of specialization: Software and Robotics.
- **Panel Moderator** — "Shaping the Future: AI Role in Transforming Digital Economies," Saudi Conference on Information Systems (SaudiCIS 2024), King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, November 2024.
- **Local Organizer and Student Volunteer** — ACM/IEEE 18th International Conference on Model Driven Engineering Languages and Systems (MODELS 2015).
- **Member** — Saudi Club in Ottawa, Canada (2010–2013). Contributed to organizing cultural, social, and sports activities, including community soccer tournaments.
- **President** — Saudi Students Association, University of Ottawa, Canada, 2012.

Skills

Programming:	Python, Java, C/C++
Machine & deep learning:	scikit-learn, PyTorch, TensorFlow/Keras
Data analysis:	pandas, NumPy, R, MATLAB
Software engineering:	UML, Umple, JUnit, ANTLR, Xtend; model-driven development, test-driven development
Databases:	MySQL, Oracle
Version control:	Git
Research tools:	Jupyter Notebook, Google Colab, Eclipse IDE, LaTeX

Languages

Arabic:	Mothertongue
English:	Fluent